Solving Systems Without a Graph

Substitution, elimination, and choosing between them

Directions

- Name your two variables and what each one stands for before you write any equations.
- Use *substitution* when one equation is already solved for a variable; use *elimination* when the same term appears in both equations.
- Give both values with the units the story uses, and check them in both equations.

1. A school trip has 12 people in all: a adults and c children, so $a + c = 12$. There are 4 more adults than children, so $a = c + 4$. Find the number of adults and the number of children.

Answer: ________________

2. At a snack bar, 2 hot dogs and 1 drink cost \$11, so $2h + d = 11$; 2 hot dogs and 3 drinks cost \$17, so $2h + 3d = 17$. Here h is the price of a hot dog and d the price of a drink, both in dollars. Find each price.

Answer: ________________

3. A school buys p pens and n notebooks, 30 items in total, so $p + n = 30$. Pens cost \$2 each and notebooks \$5 each, and the bill was \$90, so $2p + 5n = 90$. Find how many of each were bought.

Answer: ____________________

4. A courier weighs two loads. Two large boxes and three small ones weigh 31 kg: $2l + 3s = 31$. Two large boxes and five small ones weigh 41 kg: $2l + 5s = 41$. Here l and s are the masses in kilograms of one large and one small box. Find each mass.

Answer: ____________________

5. At a cinema, 4 adult tickets and 3 child tickets cost \$71, so $4a + 3c = 71$; 2 adult tickets and 3 child tickets cost \$49, so $2a + 3c = 49$. Prices are in dollars.

(a) Find the price of an adult ticket and of a child ticket.

Answer: ____________________

(b) Explain which method you used and why it was the quicker one here.

Answer: ____________________

6. A carpenter cuts a 26-metre board into two pieces of lengths x and y metres, so $x + y = 26$. The longer piece is 2 metres more than twice the shorter one, so $y = 2x + 2$. Find the length of each piece.

Answer: ____________________

7. At a bakery, 3 bagels and 2 muffins cost \$17, so $3b + 2m = 17$; 2 bagels and 5 muffins cost \$26, so $2b + 5m = 26$. Prices are in dollars. Find the price of a bagel and the price of a muffin.

Answer: ____________________

8. Gym P charges $y = 12x + 30$ dollars for x months of membership; gym Q charges $y = 18x$ dollars.

(a) Find the (months, cost) pair at which the two gyms charge the same.

Answer: ____________________

(b) Explain which method you used and why it suits this pair of equations.

Answer: ____________________

9. A caterer quotes two orders. Two trays of fruit and three trays of bread cost \$46, so $2f + 3b = 46$. Four trays of fruit and six trays of bread cost \$100, so $4f + 6b = 100$. Prices are in dollars per tray.

(a) Solve the system, or state that it has no solution.

Answer: ________________

(b) Explain what your answer means about the caterer's two quotes.

Answer: ________________

10. A boat travels on a river. Going downstream the current helps it, and it covers 20 km in one hour, so $b + c = 20$. Going upstream the current holds it back, and it covers 12 km in one hour, so $b - c = 12$. Here b is the boat's own speed and c the current's speed, both in kilometres per hour.

(a) Find the boat's speed and the current's speed.

Answer: ________________

(b) Explain what makes adding the two equations a good first move here.

Answer: ________________